Established – 1961 Subject: WEB

SEVA SADAN'S

R. K. TALREJA COLLEGE

OF

ARTS, SCIENCE & COMMERCE

ULHASNAGAR – 421 003

CERTIFICATE

This is to certify that Mr./Ms. DEEPESH GIRISH MENON of F.Y. Information Technology (FYIT) Roll No. 2541020 has satisfactorily completed the Web Designing Mini Project entitled : PORTFOLIO WITH NAVIGATION VIA USER EVENTS. during the academic year 2025 – 2026, as a part of the practical requirement. The project work is found to be satisfactory and is approved for submission.

PROF. INCHARGE **HEAD OF DEPT**

SAHIL SHUKLA **LAKSHMI JAISWAN**

INDEX

SR. NO.	CHAPTERS	PAGE NO.
1.	INTRODUCTION	1
2.	REQUIREMENT SPECIFICATION	1
3.	SYSTEM DESIGN	2
4.	SYSTEM IMPLEMENTATION	2&3
5.	SYSTEM TESTING AND RESULTS	3
6.	FUTURE SCOPE AND CONCLUSION	3&4
7.	REFERENCES	5
8.	GLOSSARY	5

1. INTRODUCTION

This project is my Personal Portfolio Website, designed to serve as a high-impact digital presence. My primary goal was to architect a platform that not only showcases my technical repertoire and project history but also demonstrates my ability to create a polished, user-centric interface.
The site is built as a Single Page Application (SPA), leveraging a modern tech stack. By combining structured HTML5, advanced CSS3 techniques (like glassmorphism and dynamic variables), and custom JavaScript logic, I have created a performant and visually engaging experience.

2. REQUIREMENT SPECIFICATION

To ensure high performance and cross-platform compatibility, the following specifications were established:
Development Environment: * IDE: Visual Studio Code with Live Server integration.
Version Control: Git/GitHub for source code management.
System Capabilities: * Compatibility: Fully optimized for Evergreen browsers (Chrome, Firefox, Safari, Edge).
Responsiveness: Designed for breakpoints ranging from mobile handsets to ultra-wide monitors.
Technologies: * Core: HTML5, CSS3, JavaScript (ES6+).
Design Patterns: Flexbox, CSS Grid, and Event-Driven Programming

3. SYSTEM DESIGN

I designed the system architecture to be modular and easily maintainable.
Dynamic Theme Engine: I implemented a logic-driven theme switcher that utilizes CSS Custom Properties to swap global color palettes instantly without a page refresh.
Navigation Architecture: A custom-built sticky header that uses a scroll-listening algorithm to apply backdrop-blur effects and transparency adjustments.
Layout Strategy:
Hero Section: A minimalist entry point featuring a custom typewriter algorithm.
Interactive Skills: A grid-based component system for clear technical categorization.
Project Showcase: Interactive cards that serve as entry points to my external repositories.

4. SYSTEM IMPLEMENTATION

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width,
initial-scale=1.0" />
    <title>Deepesh | Portfolio</title>
```

```
<style>
    :root {
        --bg-color: #0b0e14;
        --card-bg: #161b22;
        --accent: #00d2ff;
        --accent-glow: rgba(0, 210, 255, 0.3);
        --text-main: #ffffff;
        --text-dim: #94a3b8;
        --nav-bg: rgba(11, 14, 20, 0.9);
    }

    .light-mode {
        --bg-color: #f8fafc;
        --card-bg: #ffffff;
        --accent: #0ea5e9;
        --accent-glow: rgba(14, 165, 233, 0.2);
        --text-main: #0f172a;
        --text-dim: #475569;
        --nav-bg: rgba(248, 250, 252, 0.9);
    }

    * { box-sizing: border-box; transition: background 0.4s
ease, color 0.4s ease, transform 0.3s ease; }
    html { scroll-behavior: smooth; }
```

```
body {
    margin: 0;
    font-family: 'Inter', system-ui, sans-serif;
    background-color: var(--bg-color);
    color: var(--text-main);
    display: flex;
    flex-direction: column;
    align-items: center;
}

/* NAVIGATION */
nav {
    position: fixed;
    top: 0; width: 100%;
    z-index: 1000;
    padding: 20px 0;
    background: transparent;
}

.nav-scrolled {
    background: var(--nav-bg);
    padding: 10px 0;
    border-bottom: 1px solid rgba(0, 0, 0, 0.05);
    backdrop-filter: blur(10px);
}
```

```
.nav-container {
    max-width: 1000px;
    margin: auto;
    padding: 0 5%;
    display: flex;
    justify-content: space-between;
    align-items: center;
}

.logo { font-size: 1.2rem; font-weight: 800; color: var(--accent); text-decoration: none; }
.nav-right { display: flex; align-items: center; gap: 20px; }
.nav-links a { color: var(--text-dim); text-decoration: none; margin-left: 25px; font-size: 0.9rem; }
.nav-links a:hover { color: var(--accent); }

#theme-toggle {
    background: var(--card-bg);
    border: 1px solid var(--accent);
    color: var(--accent);
    width: 40px;
    height: 40px;
    border-radius: 50%;
```

```
    cursor: pointer;
    display: flex;
    align-items: center;
    justify-content: center;
    font-size: 1.2rem;
    outline: none;
}

.container { width: 90%; max-width: 1000px; padding: 120px 0 60px; }

section { padding: 60px 0; scroll-margin-top: 80px; }

header { border-left: 4px solid var(--accent); padding-left: 25px; margin-bottom: 60px; min-height: 150px; }
h1 { margin: 0; font-size: clamp(2.5rem, 8vw, 4rem); }
h1 span { color: var(--accent); border-right: 3px solid var(--accent); padding-right: 5px; animation: blink 0.7s infinite; }

@keyframes blink { 50% { border-color: transparent; } }

/* ABOUT GLOW */
.glow-text { color: var(--text-dim); line-height: 1.7; transition: all 0.5s ease; cursor: default; }
```

```
.active-glow { color: var(--text-main) !important; text-shadow: 0 0 15px var(--accent), 0 0 30px var(--accent-glow); }

/* SKILLS & PROJECTS */
.skills-grid { display: flex; flex-wrap: wrap; gap: 12px; margin-top: 25px; }
.skill-card { background: var(--card-bg); padding: 15px 25px; border-radius: 10px; border: 1px solid rgba(128, 128, 128, 0.2); cursor: pointer; }

.project-grid { display: grid; grid-template-columns: repeat(auto-fit, minmax(280px, 1fr)); gap: 25px; margin-top: 30px; }

.project-link { text-decoration: none; color: inherit; display: block; }
.project-card {
    background: var(--card-bg);
    padding: 30px;
    border-radius: 16px;
    border: 1px solid rgba(128, 128, 128, 0.1);
    cursor: pointer;
    height: 100%;
    display: flex;
    flex-direction: column;
```

```
        justify-content: space-between;
    }
    .project-card h3 { color: var(--accent); margin-top: 0; }
    .project-card p { color: var(--text-dim); font-size:
0.9rem; line-height: 1.5; margin-bottom: 20px; }

    /* GitHub Button Style */
    .github-btn {
        background: transparent;
        color: var(--accent);
        border: 1px solid var(--accent);
        padding: 8px 15px;
        border-radius: 6px;
        font-size: 0.85rem;
        font-weight: bold;
        display: inline-flex;
        align-items: center;
        gap: 8px;
        width: fit-content;
        transition: all 0.3s ease;
    }
    .project-card:hover .github-btn {
        background: var(--accent);
        color: #000;
    }
```

```
/* CONTACT FORM */
.contact-box { background: var(--card-bg); padding: 40px; border-radius: 20px; border: 1px solid var(--accent-glow); }
form { display: flex; flex-direction: column; gap: 20px; }
.form-group { display: flex; flex-direction: column; gap: 8px; }
label { color: var(--accent); font-weight: bold; font-size: 0.9rem; }
input { padding: 12px; background: var(--bg-color); border: 1px solid rgba(128, 128, 128, 0.3); border-radius: 8px; color: var(--text-main); outline: none; }
input:focus { border-color: var(--accent); }
.submit-btn { padding: 12px; background: var(--accent); color: #fff; border: none; border-radius: 8px; font-weight: bold; cursor: pointer; }

footer { width: 100%; padding: 60px 0; text-align: center; border-top: 1px solid rgba(128, 128, 128, 0.1); }
.footer-links { display: flex; justify-content: center; gap: 30px; margin-bottom: 20px; }
.footer-links a { color: var(--text-dim); text-decoration: none; font-size: 1.1rem; }
  </style>
</head>
```

```
<body>

    <nav id="main-nav">
        <div class="nav-container">
            <a href="#" class="logo">D.</a>
            <div class="nav-right">
                <div class="nav-links">
                    <a href="#about">About</a>
                    <a href="#skill">Skills</a>
                    <a href="#projects">Projects</a>
                    <a href="#contact">Contact</a>
                </div>
                <button id="theme-toggle" title="Toggle
Theme">🌙</button>
            </div>
        </div>
    </nav>

    <div class="container">

        <header id="home">
            <h1 id="typewriter-name">Hi, I'm
<span></span></h1>
            <p id="typewriter-bio" style="color: var(--text-
dim); font-family: monospace;"></p>
```

```
    </header>

    <section id="about">
        <h2>About Me</h2>
        <p id="about-text" class="glow-text">
            I am <strong>Deepesh</strong>, an 18-year-old IT student. I specialize in turning code into
            interactive user experiences using <strong>JavaScript events</strong>. I'm passionate about
            writing clean, efficient code that powers meaningful digital experiences and brings static
            designs to life.
        </p>
    </section>

    <section id="skill">
        <h2>My Skills</h2>
        <div class="skills-grid">
            <div class="skill-card">HTML5</div>
            <div class="skill-card">CSS3</div>
            <div class="skill-card">JavaScript</div>
            <div class="skill-card">React</div>
            <div class="skill-card">Node.js</div>
        </div>
    </section>
```

```
<section id="projects">
    <h2>Featured Projects</h2>
    <div class="project-grid">
        <a
href="https://github.com/Deepeshmenon666/deepesh/tree/c
0be6c52e158d2322b75f095f34d166c48898f4a/patient%20m
anagement%20system/patient%20management%20system
" class="project-link" target="_blank">
            <div class="project-card">
                <div>
                    <h3>Patient Management System</h3>
                    <p>A comprehensive system designed to
streamline healthcare workflows and manage patient
records efficiently.</p>
                </div>
                <div class="github-btn">View on GitHub
↗</div>
            </div>
        </a>


        <a
href="https://github.com/Deepeshmenon666/deepesh/blob/
24535b3ba2125c9c85671d2de580e8fc0ae3fc05/deepeshoops/
library.cpp" class="project-link" target="_blank">
            <div class="project-card">
                <div>
```

```
                    <h3>Virtual Library Assistant</h3>
                    <p>A smart OOPS-based application built
in C++ to help organize, search, and manage library
inventory digitally.</p>
                </div>
                <div class="github-btn">View on GitHub
↗</div>
            </div>
        </a>
    </div>
</section>

<section id="contact">
    <div class="contact-box">
        <h2>Let's Connect</h2>
        <form id="contact-form">
            <div class="form-group">
                <label>Email Address</label>
                <input type="email" id="user-email"
placeholder="example@email.com" required>
            </div>
            <div class="form-group">
                <label>Phone Number</label>
                <input type="tel" id="user-phone"
placeholder="+91 00000 00000" required>
            </div>
```

```
                <button type="submit" class="submit-
btn">Submit Information</button>
            </form>
            <p id="response-msg" style="text-align: center;
margin-top: 20px;"></p>
        </div>
    </section>

</div>

<footer>
    <div class="footer-links">
        <a href="https://github.com/Deepeshmenon666"
target="_blank">GitHub</a>
        <a href="#">LinkedIn</a>
        <a href="#">Instagram</a>
    </div>
    <p style="color: var(--text-dim);">&copy; 2026
Deepesh. Built with passion & code.</p>
</footer>

<script>
    // EVENT 1: Typewriter Effect
    const nameText = "Deepesh";
    const bioText = "> IT Student & Aspiring
Developer_";
```

```
let nameIdx = 0;
let bioIdx = 0;

function typeEffect() {
    if (nameIdx < nameText.length) {
        document.querySelector('#typewriter-name span').textContent += nameText.charAt(nameIdx);
        nameIdx++;
        setTimeout(typeEffect, 150);
    } else if (bioIdx < bioText.length) {
        document.getElementById('typewriter-bio').textContent += bioText.charAt(bioIdx);
        bioIdx++;
        setTimeout(typeEffect, 50);
    }
}
window.onload = typeEffect;

// EVENT 2: Theme Toggle
const themeBtn = document.getElementById('theme-toggle');
themeBtn.addEventListener('click', () => {
    document.body.classList.toggle('light-mode');
```

```
        themeBtn.innerText = document.body.classList.contains('light-mode') ? "☼" : "☾";
    });

    // EVENT 3: Scroll Navbar
    window.addEventListener('scroll', () => {
        const nav = document.getElementById('main-nav');
        window.scrollY > 50 ? nav.classList.add('nav-scrolled') : nav.classList.remove('nav-scrolled');
    });

    // EVENT 4: About Text Glow
    const aboutText = document.getElementById('about-text');
    aboutText.addEventListener('mouseover', () => aboutText.classList.add('active-glow'));
    aboutText.addEventListener('mouseleave', () => aboutText.classList.remove('active-glow'));

    // EVENT 5: Skill & Project Hover Effects
    const cards = document.querySelectorAll('.skill-card, .project-card');
    cards.forEach(card => {
        card.addEventListener('mouseenter', () => {
            card.style.borderColor = 'var(--accent)';
```

```
            card.style.boxShadow = '0 10px 25px var(--accent-glow)';
            card.style.transform = card.classList.contains('skill-card') ? 'translateY(-5px)' : 'translateY(-10px)';
        });
        card.addEventListener('mouseleave', () => {
            card.style.borderColor = '';
            card.style.boxShadow = 'none';
            card.style.transform = 'translateY(0)';
        });
    });

    // EVENT 6: Form Submit
    const form = document.getElementById('contact-form');
    form.addEventListener('submit', (e) => {
        e.preventDefault();
        const email = document.getElementById('user-email').value;
        const msg = document.getElementById('response-msg');
        msg.style.color = 'var(--accent)';
        msg.innerText = "Success! Deepesh will contact you at " + email;
        form.reset();
```

```
    });

    // EVENT 7: KEYBOARD SHORTCUTS
    window.addEventListener('keydown', (e) => {
        if (e.ctrlKey) {
            const key = e.key.toLowerCase();
            let targetId = "";

            if (key === 'a') targetId = "about";
            if (key === 's') targetId = "skill";
            if (key === 'p') targetId = "projects";
            if (key === 'c') targetId = "contact";

            if (targetId) {
                e.preventDefault();
                const targetElement =
document.getElementById(targetId);
                if (targetElement) {
                    targetElement.scrollIntoView({ behavior:
'smooth' });
                }
            }
        }
    });
  </script>
```

```
</body>
</html>
```

5. SYSTEM TESTING AND RESULTS

Testing is a crucial phase of the development lifecycle to ensure that the website functions as intended across different environments and user interactions. I conducted Functional Testing, UI/UX Testing, and Compatibility Testing to validate the portfolio's performance.

5.1 Test Cases and Observations

The following table outlines the specific test cases executed, the expected outcomes, and the final results

Test ID Feature Tested Action Performed Expected Result Status

TC-01 Typewriter Effect Page Load Name and bio should animate text character by character. Pass

TC-02 Theme Toggle Click the / icon UI colors should swap via CSS variables; icon should update. Pass TC-03 Sticky Navigation Scroll down the page Navbar should gain a blurred background and border for visibility.

Pass

TC-04 Smooth Scrolling Click "Projects" in Nav Page should scroll smoothly to the target section anchor. Pass

TC-05 Hover Interactivity Mouse over Skill/Project card Card should scale up slightly and display a glow effect. Pass

TC-06 Form Validation Click "Submit" with empty fields Browser should prevent submission and prompt for required fields. Pass

5.2 Testing Summary

Logic Verification: All JavaScript event listeners (Scroll, Click, Keydown) responded within the expected millisecond range, ensuring no lag in user interaction.
Visual Consistency: The use of CSS variables ensured that the theme transition was seamless across all sections without any "flash" of unstyled content.
Cross-Browser Results: Tested on Chrome, Firefox, and Safari. The Intersection Observer and Flexbox/Grid layouts remained consistent across all modern rendering engines.
Note: All tests were conducted in a local development environment and verified through browser developer tools to ensure console-free errors during execution.

6.FUTURE SCOPE AND CONCLUSION

Future Scope
Headless CMS: I plan to integrate a CMS (like Sanity or Strapi) to manage project content dynamically.
Serverless Contact Form: Moving from local state feedback to an API-based email notification system.
Motion Libraries: Incorporating advanced physics-based animations for smoother transitions

7.REFERENCES

The development of this portfolio was supported by the following technical resources and documentation: MDN Web Docs (Mozilla Developer Network): Utilized for comprehensive documentation on JavaScript Event Listeners, the Intersection Observer API, and semantic HTML5 tags.

W3Schools Online Web Tutorials: Referenced for CSS Flexbox and Grid layout properties to ensure a responsive design.
Google Fonts & Icons: Sourced the 'Inter' typeface and utilized Unicode character sets for UI iconography.
Canva / Figma: Used for initial wireframing and color palette selection (Dark/Light mode contrast ratios). Stack Overflow: Consulted for best practices regarding keyboard event handling (keydown) and preventDefault behaviors.

8.GLOSSARY

SPA: Single Page Application; a web app that loads a single HTML page and dynamically updates content.
Event-Driven: A programming paradigm where the flow is determined by user actions (clicks, keypresses). CSS Variables: Entities defined by CSS authors that contain specific values to be reused throughout a document.event to occur (like a click).
preventDefault(): A method used to stop the default action of an element (e.g., stopping a form from refreshing the page).
Vanilla JavaScript: Plain JavaScript without any additional libraries or frameworks.